

Author Response to Reviewer 1P2E

All changes marked in blue; line refs refer to revised `main.tex`.

R1.1: Theoretical discussion (bias-variance, parameterization complexity, dataset size).

Response: Expanded simplicity principle discussion [L604--605]: explicit parameter-count analysis (~ 180 vs. ~ 7 samples/parameter for invariants vs. lowrank), connected to model selection theory. New feature dimension ablation [L484--508] (Table 6, [L489--504]): RMSE monotonically degrades $D=256 \rightarrow 2048$, confirming fundamental statistical constraint. Conclusion updated [L618].

R1.2: Classification benchmarks. **Response:** Added BACE and BBBP experiments: Section 5.5 [L461--481], Table 7 [L466--479]. Conformer features hurt BACE (-4.4% AUROC), no effect on BBBP ($+0.1\%$)—consistent with taxonomy. Scope statement [L151], abstract [L80] updated. Per-seed results in Appendix K [L1041--1064].

R1.3: Larger/industrial datasets. **Response:** Acknowledged as limitation [L620]. Benchmarks span 642–133K molecules; validation on ChEMBL/ZINC noted as future work.

R1.4: Figure resolution, standardize metrics, typos. **Response:** Figures re-exported ≥ 300 DPI. Metrics standardized: RMSE (regression), AUROC (classification) [L461--481]. Full proof-reading pass completed.

R1.5: Scaffold split generation and replicates. **Response:** Expanded Training Details [L274]: Bemis–Murcko scaffold extraction via RDKit, frequency-sorted assignment, 80/10/10 splits. Seed counts: 3 (main benchmark), 10 (statistical validation). Classification uses stratified splits [L464].